

Percentage – APTITUDE LEARNING NOTES

Complete concepts • formulas • shortcuts • solved examples • practice

1. Percentage Basics

Percentage means per hundred: $x\% = x/100$.

FORMULA: $\text{Percentage} = (\text{part}/\text{whole}) \times 100$.

Convert fraction $\rightarrow$ percentage by $\times 100$; percentage $\rightarrow$ fraction by $\div 100$.

2. Increase and Decrease

FORMULA: $\text{Percentage change} = (\text{change}/\text{original}) \times 100$.

New value after $r\%$ increase = $\text{original} \times (1 + r/100)$.

New value after $r\%$ decrease = $\text{original} \times (1 - r/100)$.

3. Successive Changes

FORMULA: $\text{Net multiplier} = (1 \pm a/100)(1 \pm b/100) \dots$

Two increases $a\%, b\% \rightarrow$ net increase = $a + b + ab/100\%$.

Increase $a\%$ then decrease $b\% \rightarrow$ net change = $a - b - ab/100\%$.

4. Reverse Percentage

FORMULA: $\text{Original after } r\% \text{ increase} = \text{final} / (1 + r/100)$.

FORMULA: $\text{Original after } r\% \text{ decrease} = \text{final} / (1 - r/100)$.

5. Profit, Loss and Discount

FORMULA: $\text{Profit} = \text{SP} - \text{CP}$; $\text{Loss} = \text{CP} - \text{SP}$.

FORMULA: $\text{Profit}\% = \text{Profit}/\text{CP} \times 100$; $\text{Loss}\% = \text{Loss}/\text{CP} \times 100$.

FORMULA: $\text{Discount} = \text{MP} - \text{SP}$; $\text{Discount}\% = \text{Discount}/\text{MP} \times 100$.

6. Price, Consumption and Expenditure

If price rises $r\%$ and expenditure stays fixed, consumption must fall.

FORMULA: $\text{Consumption decrease} = 100r / (100 + r)\%$.

If price falls $r\%$, required consumption increase = $100r / (100 - r)\%$.

7. Marks, Population and Common Conversions

$1/2 = 50\%$, $1/4 = 25\%$, $1/5 = 20\%$, $1/8 = 12.5\%$, $1/10 = 10\%$, $1/20 = 5\%$, $1/25 = 4\%$.

Population, salary, marks, production etc. are solved with the same percentage multiplier.

Solved Aptitude Examples

Q1. Find 25% of 360.

Solution: $360/4=90$.

Q2. 500 becomes 575. Increase?

Solution: $75/500 \times 100 = 15\%$.

Q3. After 20% decrease value is 640. Original?

Solution: $640/0.8=800$.

Q4. Increase 20% then decrease 10%.

Solution: $1.2 \times 0.9 = 1.08 \rightarrow 8\% \text{ increase}$.

Q5. CP ■800, SP ■920. Profit%?

Solution: $120/800 \times 100 = 15\%$.

Q6. MP ■2000, discount 15%. SP?

Solution: $2000 \times 0.85 = \text{■}1700$.

Q7. Population 50,000 rises 12%.

Solution: **56,000**.

Q8. Price rises 25%; same expenditure. Consumption fall?

Solution: $100 \times 25/125 = 20\%$.

Q9. 72 marks out of 90.

Solution: **80%**.

Q10. 20% discount price is ■960. MP?

Solution: $960/0.8 = \text{■}1200$.

Practice Questions

1. Find 35% of 480.

2. 800 becomes 920. Find increase %.

3. After 25% decrease value is 450. Find original.

4. Increase 10% then 20%. Net increase?

5. Decrease 20% then 25%. Net decrease?

6. CP ■1200, SP ■1380. Profit%.

7. MP ■2500, discount 18%. SP?

8. Population 80,000 falls 7.5%. New population?

9. Price rises 20%. Consumption decrease for same expenditure?

10. Price falls 20%. Consumption increase for same expenditure?

11. 72 is what percent of 240?

12. 360/450 is what percentage?

13. Increase 30% then decrease 30%. Net change?

14. After 15% discount price is ■1700. MP?

15. 40% of a number is 96. Find number.

Answer Key

1) 168 • 2) 15% • 3) 600 • 4) 32% • 5) 40% • 6) 15% • 7) ■2050 • 8) 74,000 • 9) 16■% • 10) 25% • 11) 30% • 12) 80% • 13) 9% decrease • 14) ■2000 • 15) 240

Quick Revision Formula Sheet

- Percentage=(part/whole)×100
- Increase r% → $\times(1+r/100)$
- Decrease r% → $\times(1-r/100)$
- Successive changes → multiply factors
- Profit%=profit/CP×100
- Discount%=discount/MP×100
- Price rise r% → consumption fall= $100r/(100+r)\%$

Exam strategy: Identify the pattern first, choose the shortest valid formula, solve step-by-step, and verify the final answer.